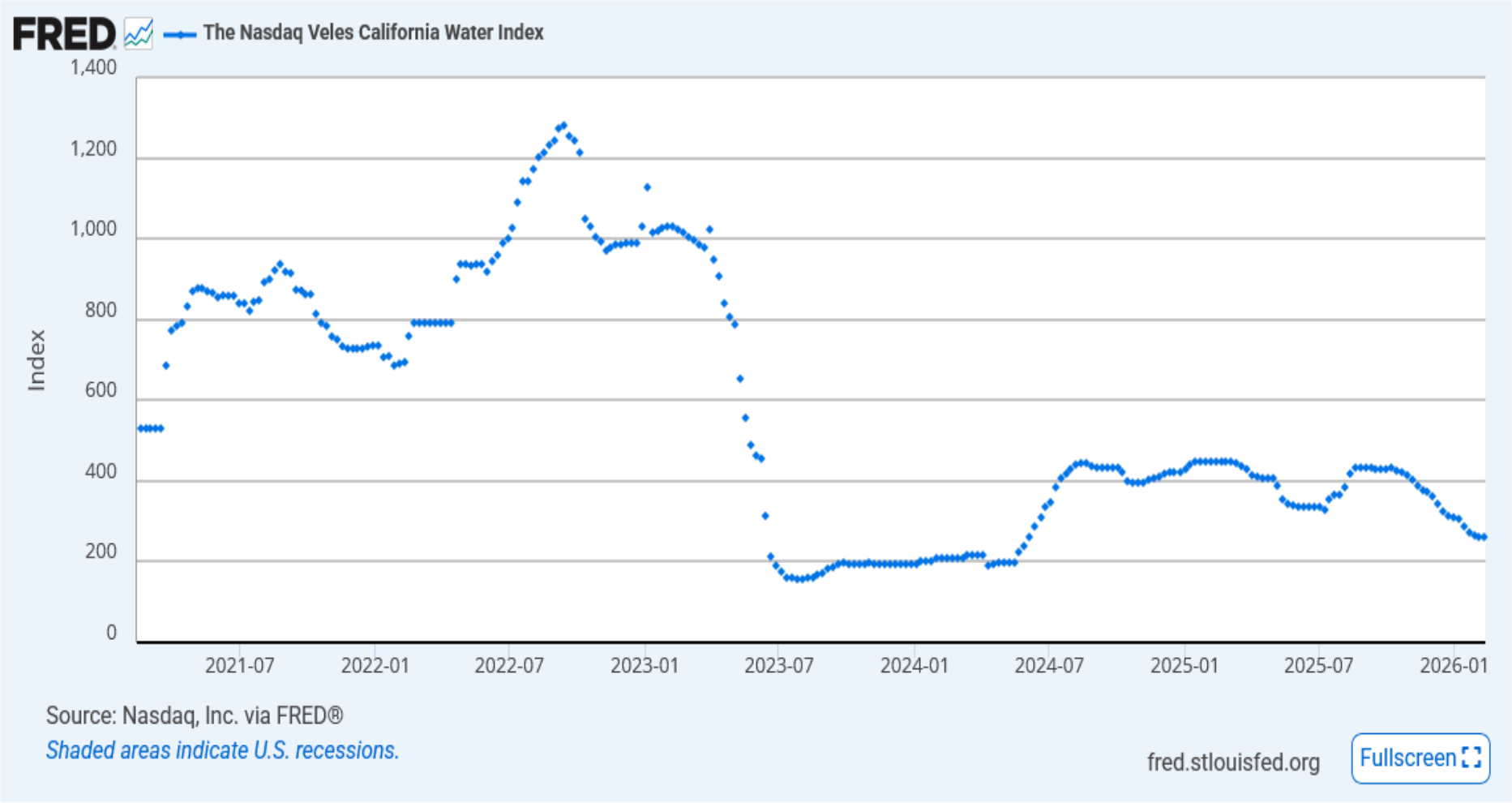
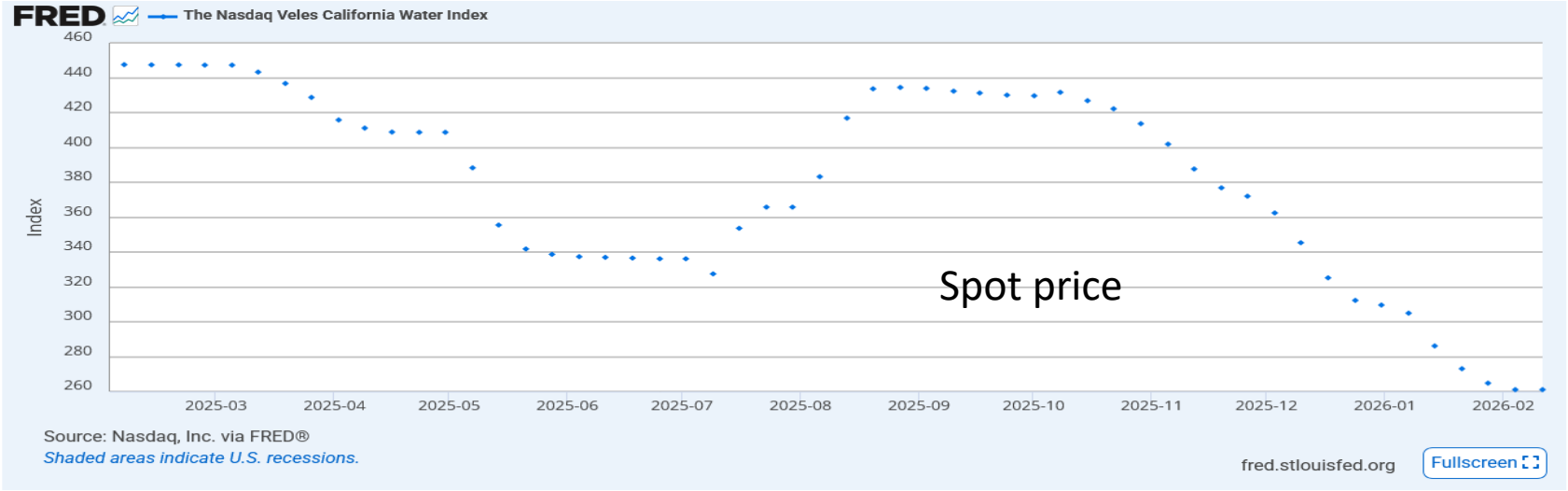
Water Spot prices



Future contract price for delivery June 2026
price of acre foot of water
why no impact of current rainfall?



Class 17
Forwards and Futures
Derivatives

- **Definitions**
  - **Instruments**
  - **Markets**
- **Financial markets as insurance markets**
- **The counterparty risk problem**
- **Markets as solutions**
- **Hedging**

Key concept

- Forward vs Future
- Strike price
- Arbitrage constraint
- Margin account
- Hedging

Contracts so far

- Stock/Equity
  - Fraction of the residual value of the firm (after all other payments)
- Bonds/Debt
  - Promise to pay.
- Claims on cash flows, uncertain but not contingent.
- Trade in the stocks and debt is ‘instantaneous.’
  - Buy or sell a claim pay now.
- What about contract now pay later?
- What about contingent claims?

Other contracts

- Forwards
  - Contract now for later execution
- Contingent claims
  - Insurance
  - options
- Common features.
  - Execution risk
  - Associated with some other asset.
- Derivatives
  - Contract over financial assets

Forwards

- Forwards
  - Promise to deliver at a given price paired with promise to buy
  - Example: Solar Field Operator promises to deliver 8GWh a day at 3c a watt-hour for 30 years & LADWP promises to buy.
  - Example 2, Poultry farm promises to deliver 200K dozen eggs (@72 cents) for delivery in 90 days. Fast foot chain promises to purchase
- Definition
  - Notional the quantity of the contract
    - (8GWh a day); (200,000 dozen);
  - Underlying: the item at stake here
    - (electricity) (eggs)
  - Delivery: the date the trade is to be completed
    - (over thirty years) (90 days)
  - Short position: promise to sell
  - Long position: promise to buy

Forwards

- A forward contract is a promise to purchase a specific quantity of some item at a specified date.
  - These are common in B2B environments where both the buyer and the seller may want to stabilize delivery and prices (say a laptop manufacturer and a hard drive manufacturer)
  - Commercial Aircrafts, government contracts…
- Over the counter (privately agreed)
  - Firm (not option)
  - Details (price, quantity, maturity,…) decided for each contract
  - Payment is due upon delivery.
  - Illiquid

Derivatives

- Last page notional: Eggs, aircraft.
- Derivatives are contracts whose notional are other financial contracts
  - Single stock futures
  - Index futures
  - Options
  - …

Market based forwards

- Futures
  - promise to buy or sell at specified prices
  - Commodities
  - Derivatives (S&P 500)
- Options
  - Right to buy or sell
  - Commodities
  - Buy also financial assets
- Designed to deal with risk, also present arbitrage opportunities

Price of S&P 500 future that settles Thursday

03/19/2026

More forwards

- Swaps
  - Right to exchange one asset for another.
    - The right to exchange 150 shares of Apple for X shares of AMD over the next 5 years.
    - Swap arise only if AMD rises faster than Apple
  - Interest rate swap
    - Exchange my interest payments (6.75%) on 1 million dollars for the Fed Funds rate rate +3%.
    - Buyer benefits if FF rate falls. Seller benefits if FF rate rises.
  - Credit default swap.
    - Contingent contract. The seller agrees to replace a defaulted bond rate BB+ with a non-defaulted BB+ bond of similar yield and maturity today.

Forwards basics

- Transaction agreed today at for price in the future (p_s) the strike price
  - p_s is not p_0 the price at the time the deal was struck
- At the time the contract comes due the price (p_t) will be different (almost for sure) from p_s
- If $p_s > p_t$ the buyer would rather not execute
  - (pay less on the open market)
- If $p_s < p_t$ the seller would rather not execute
  - (get more on the open market)
  - More one side or the other may not be able to execute (Boeing when 737Max was grounded)
- So we have execution risk

Linking current and future prices

- Two arbitrage condition
- Long position
  - SL1 Hold the asset+go short on futures
    - Cost: Forgone interest + storage
  - SL2 Sell the asset now.
- Short position
  - SS1: go long on future
  - SS2: buy now
    - Cost: Forgone interest + storage
- Price of future have to be within the two arbitrage bounds

Forwards' and Futures' prices

- Back to the wheat futures.
- Take a baker who wants to insure the price of flour say 6 months in advance.
- Two choices buy ahead and store, or buy futures
  - In equilibrium baker must be indifferent.
- Storing can be either costly or not.
  - In the case of wheat have to keep it somewhere
  - Have to pay for the wheat (so there is forgone income)
  - The price today (p_0) and (F^T) the price for delivery at T are connected by an arbitrage condition
- $F^T = (1+r+c)^T p_0$ where r is the interest income foregone and c is the per unit storage cost. (pay r+c each period)

Arbitrage on the long side
(own the asset now, intend to sell in T period)

- Future (F^T) and the current spot price (p_0) are connected
- If storage costs are negligible (diamonds)
  - Taking the long position and going short on futures
    - foregone capital income
  - Short-term interest rate (r) => $F^T = (1+r)^T p_0$
- If there is a per period storage cost (oil, bacon)
  - Storage cost c => $F^T \approx (1+r+c)^T p_0$
- Financials with a per-period return (dividends, interest)
  - Income y => $F^T \approx (1+Tr-y)^T p_0$

Arbitrage on the short side
(do not own the asset now, intend to buy in period T)

- The cost are on the buy now strategy
- No cost to going long on futures
  - Short-term interest rate (r) => $F^T = (1+r)^T p_0$
  - Storage cost c => $F^T \approx (1+r+c)^T p_0$
  - Income y => $F^T \approx (1+r-y)^T p_0$

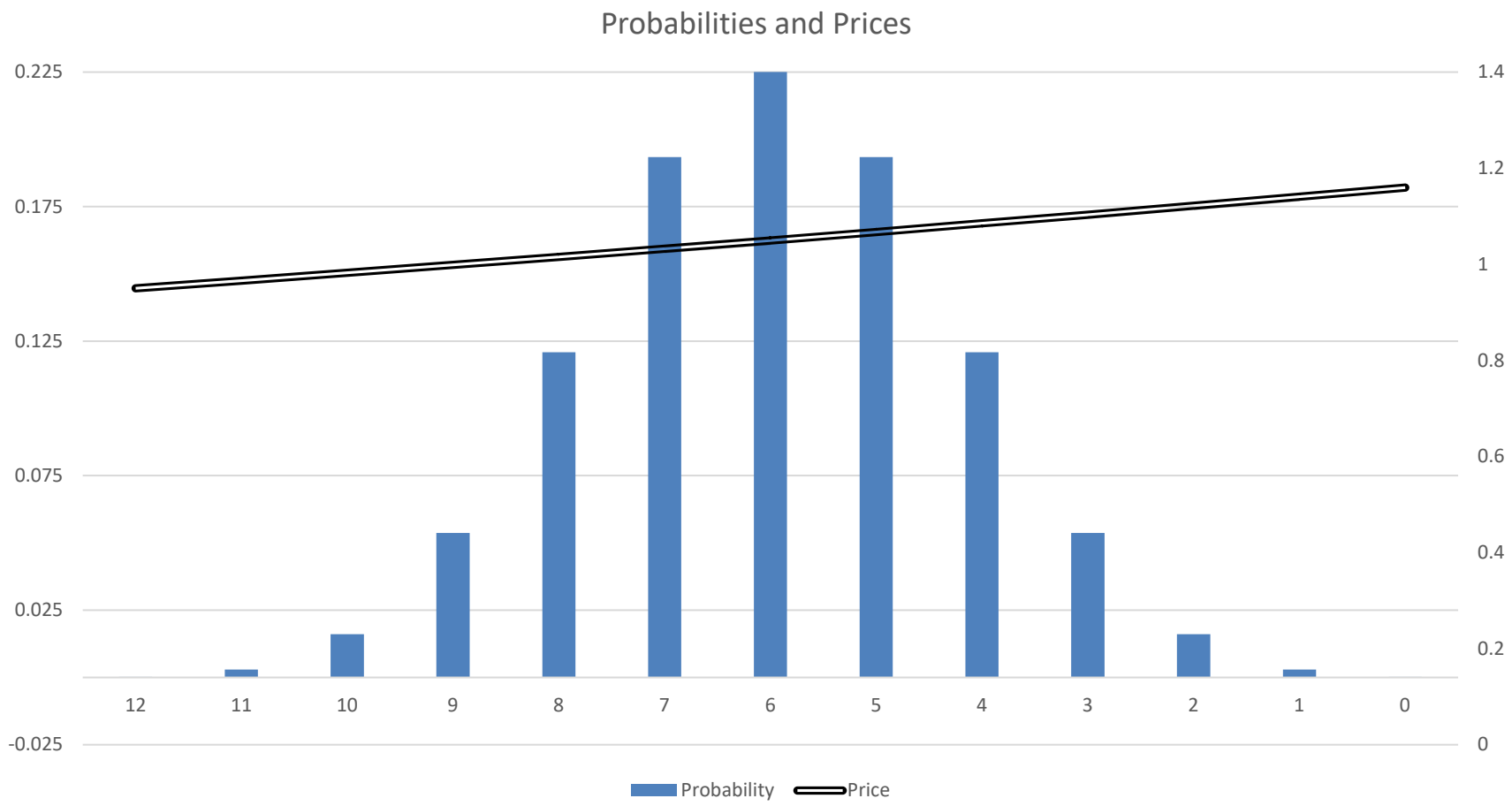
  So the arbitrage conditions are the same

Counter party risk

- Identity of the counterparty really matters.
  - On both side
  - Buyer does not want to order planes if not delivered
  - Seller does want to produce planes that no one buyers.
- When you sign a long-term contract with a cell phone company you are in effect signing a forward (buying X minutes a month at some pre-specified price).
- Often there are escape clauses

Modeling forwards

- Back to our simple binomial distribution. (VAR/CVAR)
- Model the possible range of prices 12 months hence. Each month the price may go up by 1.2% or down by 0.43% so the expected monthly return is 0.04% and the annual rate is 5%
- The current price is 1 and expected value of our security is 1.05 if everyone is risk-neutral p_s=1.05
- What is the distribution of possible prices?



Horizontal axis is number of down draws

Forward contracts as zero-sum game

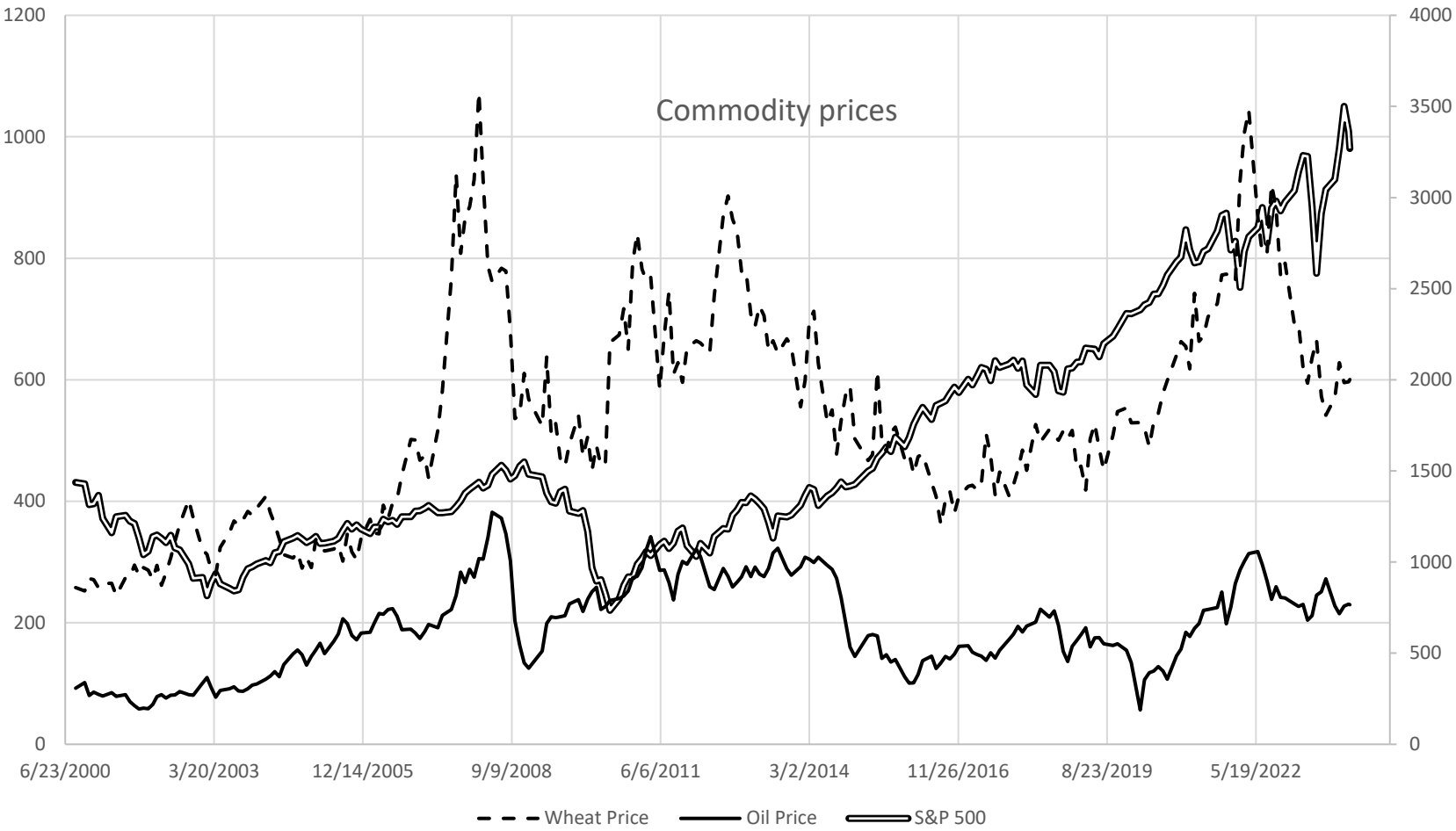
- From before p
  - Suppose price appreciation has been high after a year (8 good draws) p_t=1.08
    - Buyer gets item at p_s=1.05 can sell on the open market at p_t=1.08
    - Profit is .03 times the notional (the capital value of contract)
    - Seller loses the same amount
  - Suppose price depreciation has been high after a year. (9 bad draws) p_t=0.998
    - Buyer gets item at p_t=1.05 could have bought on the open market at 0.998
    - Loss to buyer is 0.0502 times the notional
    - Seller makes a profit of the same amount
- More generally
  - Buyer's gains is seller's loses
  - $\Pi_b = p_T - p_s \quad \Pi_S = p_s - p_T \Leftrightarrow \Pi_b = -\Pi_S$
  - That is a zero-sum game. And incentives to pay up have to be low for one side or the other, this is particularly true if $p_T - p_S$ is large
  - Need an enforcement mechanism

Problem and Solution

- Courts
  - For reasons not clear, there has been a lot of suspicion about forward markets in the past
  - Courts have tried to suppress them or refused to enforce such contracts.
  - Then after the 1850s more enforcement
- Repeated dealings
  - Small number of players who contract repeatedly, if these are fair bargains then, you can secure cooperation (people pay up) as long as $p_T - p_0$ is not too large. (this involve forwards being relative short durations).
- Trading mechanism

Power of small jumps

- We start with a contract 12 months 1.05
- After 1 month the price is either 1.012 or 0.9957
- Recompute the expected value 11 months later
- If the first month was up $E(p_t)=1.0587$
- If the first month was down $E(p_t)=1.02368$
- We can keep the contract balance by having the buyer and seller deposit the expected change 0.0087 at start
  - When first month is determined the change in price is transferred from one account to the other.
  - The looser has to do the same for period 2 (add cash to her account)
- Claim this restores balance and that is what futures markets do.



	Oil	S&P 500
Wheat	0.77	0.345
Oil		0.099

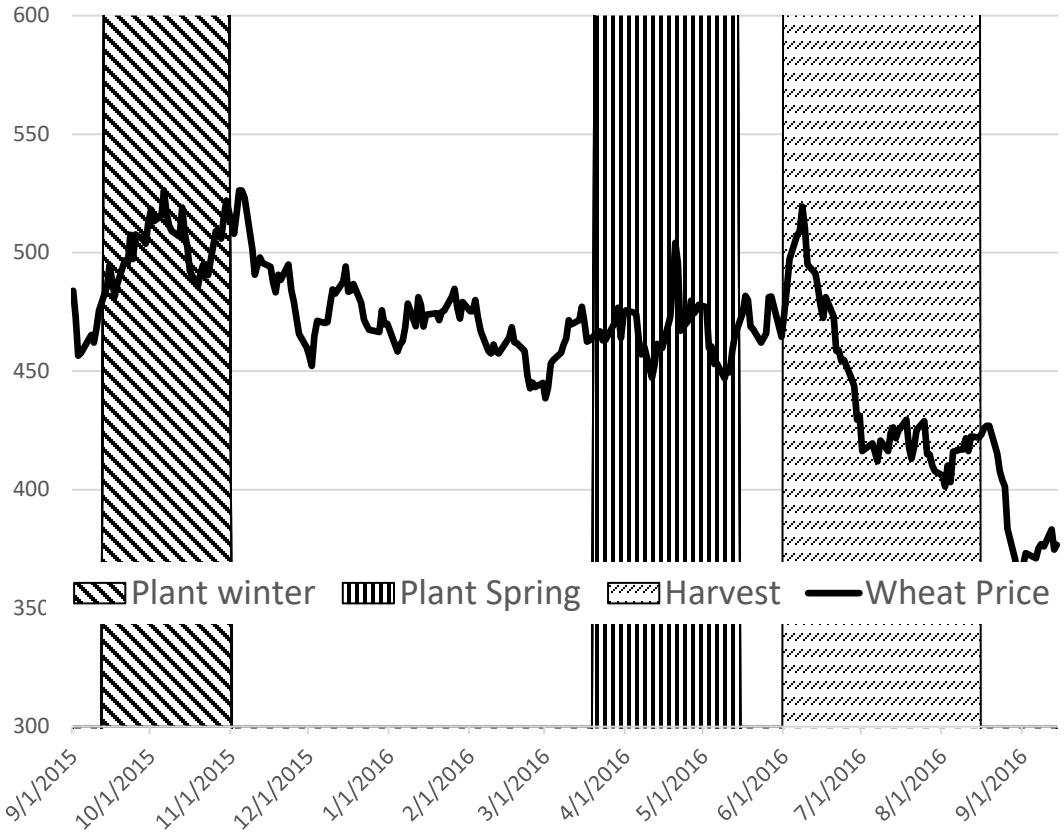
Commodities are good short-term sources of diversification (low correlation with equity prices)

Bad long-term investments (output is related to price)

So why so many futures

The farmers' problem

- Decide what to plant and when to plant
- Engage in cost
- Payoff uncertain.
- Harvest occurs
  - Two sources of uncertainty quantity
  - Price
  - Use futures to reduce risk (eliminate the price uncertainty)
  - Take a short position to offset their long position in wheat at harvest.
  - But only on part of output



Problem arises for firms that are dependent on inputs with volatile prices or face volatile output markets

Implementing futures

- A firm wants a long position in some commodity
  - Start by opening a collateral account
- Contract is initiated when some seller accepts the firm's bid.
  - The collateral account must have a balance of at least 10% of the value of the contract
- The exchange then initiates two contracts.
  - One between itself and the buyer,
  - One between itself and the seller.
  - The two contracts offset each other so the exchange has no aggregate liability.
- Collateral account must meet the balance requirement
  - If your position improves, you can withdraw, if it worsens you must post additional collateral.
- Positions are closed by selling the contract back to the intermediary before expiration.
- Does this solve the zero-sum game?

Futures keep incentives in check

- Margin account
- Deposit has to be at least the expected change in value NEXT PERIOD
- If margin account falls below this value exchange liquidates the position immediately.